Noah Sun

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Education

University of Waterloo – BASc. in Mechatronics Engineering (Candidate) September 2025 – April 2030

Relevant Coursework: Algorithms and Data Structures, Circuits, Digital Logic GPA: 3.92 / 4.00

Experience

Full Stack Software Engineer – Thames Valley Financial (Incoming) September 2026 – December 2026

- Building full stack features for a financial administration platform, including internal tools and product interfaces
- Working AI-forward, steering agentic coding tools toward production-ready features in a small engineering team

Forward Deployed Engineer – Ascendance Foundry January 2026 – April 2026

- Developed **5 AI workflow automations**, introducing input validation and reducing human error by **80%**
- Built a GTM automation that uses Python Playwright to find leads and generate outreach emails using Claude API
- Implemented **2 ETL pipelines** using Azure Data Factory, delivering per-project statistic dashboards in Power BI

Co-founder and Lead Developer – Elapse | Google Play | App Store | GitHub June 2024 – August 2025

- Led 4 developers to build an iOS/Android Flutter app serving the **100k+** member VEX Robotics Competition community with real-time schedules, rankings, and match data, receiving **3500+ downloads** and a **5.0-star rating**
- Designed adaptive match times and scouting forms, increasing team productivity at tournaments by **35%**
- Integrated Firebase and Cloud Functions to sync data between team members, reducing scouting times by **30%**

Embedded Flight Software Developer – Waterloo Aerial Robotics Group September 2025 – Current

- Led 4 engineers to develop a **32 MB** flash driver using C++ on the STM32 Nucleo to store config details and crash logs
- Built an RTOS-integrated Flash Translation Layer with garbage collection and wear leveling, enhancing flash longevity

V5RC Team Programming Lead – Checkmate Robotics | Demo Video | GitHub June 2019 – August 2025

- Led team of 6 in robot design and library architecture, winning the Create (innovation) Award at VEX Worlds 2024
- Engineered ArkLib, a modular C++ motion control library focused on scalability and flexibility using OOP, qualifying the team to VEX Worlds in **5 consecutive years** and winning the Think (programming) Award at international events
- Designed and tuned PID and control algorithms, achieving **1”/1° precision** movements across **300+ matches**

Projects

Redact | Live Demo | GitHub

- Built a full-stack face-blurring app (FastAPI, React, Docker) combining RetinaFace detection with OpenCV tracking
- Achieved **97% accuracy** by detecting every 5 frames with IoU-tracked fill-in, cutting processing times by **80%** to **4 FPS**
- Designed dual-signal scene-cut detection and EMA smoothing to keep blur boxes stable across shot changes

Signal - LLM Agent Simulation Platform | GitHub

- Built a multi-layer LLM agent simulation and dashboard in Python and JavaScript with clean separation across a simulation kernel, Claude tool-use harness, and FastAPI/SSE server
- Implemented structured tool-use over Claude API with prompt caching, reducing per-step latency and token cost

CityWatch | GitHub

- Led a team of 4 to build a reporting platform using React Native to analyze **50k+ reports** to identify high-risk zones
- Trained a MobileNetV2 road defect classifier with OpenCV preprocessing pipeline, achieving **95%+ accuracy**
- Forecasted 30-365 day risk heatmaps from spatial incident patterns and features stored in a PostGIS database

Technical Skills

Languages: C/C++, Python, Java, JavaScript, TypeScript, Dart
Frameworks/AI: ROS2, OpenCV, TensorFlow, Claude API, FastAPI, Express.js, Node.js, React Native, Flutter, Matplotlib
Data/Backend: PostgreSQL, MySQL, Microsoft Azure, Firebase, Pandas
Developer Tools: Git, Docker, Linux, GitHub Actions, Foxglove, REST APIs, SSE, Microsoft Power Apps, MS Office
Hardware: Raspberry Pi, STM32 Nucleo, NVIDIA Jetson Nano, Arduino